CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
6th Semester of B. Pharm. Examination
University Theory Examination May 2017
PH324 Pharmaceutical Analysis II

Date: 12.05.2017 , Friday **Time:10:00a.m. to1:00p.m.** **Maximum Marks: 80**

Instructions:
1. **There are three sections in this question paper.**
2. **SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.**
3. **Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.**
4. **Figures to right indicate marks.**
5. **Draw neat labeled sketches wherever necessary.**

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. **20**

1. Affinity Chromatography is especially known for the the separation of
[A] Proteins
[B]Cations and Anions
[C] ligands
[D] small molecules

2. The particle size in stationary phase packing material (Column) is____ for UPLC technique
[A]< 5 microns
[B]< 2 microns
[C]> 2 microns
[D]> 5 microns

3. Requirement for a compound to show optical activity is
[A]presence of racemic mixture
[B] centre of asymmetry
[C]centre of symmetry
[D]D- isomer

4. The value of Rf ranges from
[A] 0 to 1
[B] 1 to 10
[C] 10 to 100
[D] 0 to 100

5. In Chromatography, Change in velocity of mobile phase doesnot affect___
[A]Eddy diffusion
[B]Longituditional diffusion

[C]Mass transfer
[D]All of the above

6. Decrease in particle size of stationary phase _________ the separation efficiency
[A]Decreases
[B] Increases
[C] Does not Affect
[D] May Affect

7. Which of the following adsorbent possess strongest adsorption power?
[A] Activated charcoal
[B]Silica gel
[C]Starch
[D]Calcium carbonate

8. When the potential applied across two electrode is maintained at some constant value, the current is measured and plotted against the volume of the titrant, this technique is known as
[A]Conductometry
[B]Potentiometry
[C]pH metry
[D]Amperometry

9. Which one is NOT a reference electrode
[A]Silver silver chloride
[B]Hydrogen
[C]Dropping Mercury
[D] Calomel

10. Chemical noise is generated due to the variables that affect the
[A]component of an instrument
[B]measurement system
[C] chemistry of the system being analyzed
[D]All of the above

11. ____________ is the technique used to analyze a mixture containing wide range of Boiling Points like mixture containing A = 100°, B = 150°, C = 250°, D = 300°.
[A]Precolumnderivatization
[B]Postcolumnderivatization
[C]Temperature Programming
[D]Calibration curve method

12. Which of the following statements about chromatography is correct?
[A]Paper chromatography and gas chromatography are both routinely used for qualitativeanalysis only.
[B]Paper chromatography is usually considered to be quantitative only, while gaschromatography can be qualitative or quantitative.
[C] Paper chromatography is usually considered to be qualitative only, while gas chromatography can be qualitative or quantitative.
[D]Paper chromatography and gas chromatography are both routinely used for quantitativeanalysis only.

13. An extraction system which require less solvent and produces a more concentrated extract phase, isdesired with a __________ compound

[A]large distribution coefficient

[B] small distribution coefficient

[C]very small distribution coefficient

[D] constant distribution coefficient

14. Platinum foil is coated with platinum black in which type of electrode?

[A] Hydrogen

[B] Glass

[C] Mercury

[D] Silver

15. In Polarography, any change in diffusion current is denoted by

[A]Nernst Equation

[B]Ilkovic equation

[C]Arrhenius equation

[D]Stock equation

16. Polarography can be used to

[A]Study fluorescent material

[B]Study resistance of a solution

[C] Study current potential relationship

[D]Study optical activity of organic molecules

17. Ion Exchange chromatography is based on

[A]Electrostatic attraction

[B] Mobility of ion species

[C]Absorption chromatography

[D] Size of the ions

18. Chromatography can be used to

[A]form mixtures

[B]change mixture compositions

[C]separate mixtures into pure substances

[D]all of the above

19. Separation efficiency in Gas Chromatography is inversely proportional to

[A] Internal diameter of column

[B]Outer diameter of column

[C]Length of column

[D]None of above

20. Which of the following is not a component of GC Instrument?

[A]Injection Device

[B]Temperature Control Device

[C] Degassing Unit

[D]Flow Regulator

SECTION – II

Q 2 Attempt any**TWO**of the following

A Explain Stripping Voltametry. **05**

B	Describe the steps involved in Thin Layer Chromatography.	05
C	Enlist i. Factors affecting rate and selectivity of extraction process. ii. Considerations for selection of solvent for extraction process .	05

Q 3	Attempt any **TWO** of the following	
A	Compare HPLC and UPLC.	05
B	Construct and discuss the plot of conductance versus ml of titrant for i. Hydrochloric acid titrated with Ammonium Hydroxide ii. Acetic acid titrated with Potassium Hydroxide	05
C	Discuss thetypes of GC detectors and the selection of carrier gas based on the detector used.	05

SECTION – III

Q 4	Attempt any **FOUR** of the following	
A	What is chromatography? Explain plate theory of chromatography.	05
B	Explain the principle of Column Chromatography. What are the criteria for selection of adsorbents in column chromatography? What is Silica gel GF_{254}?	05
C	Discuss different types of detectors used in HPLC and write about absorbance detector.	05
D	Give brief account on GC columns and packing materials.	05
E	How will you optimize column performance in HPLC?	05
F	Discuss the mechanism of ion exchange in Ion Exchange chromatography and enlist the requirements of a resin to be used in Ion Exchange chromatography	05

Q 5	Attempt any **FOUR** of the following	
A	Write down advantage and applications of HPTLC.	05
B	Describe various types of development techniques for Paper Chromatography.	05
C	Discuss the principle Polarimetry analysis. Discuss any pharmacopoeial application of polarimetry analysis.	05
D	Enumerate different types of reference electrodes. Explain calomel electrode in detail.	05
E	Explain principle of polarography with Ilkovic equation.	05
F	What is Kohlrausch's law. Enlist the factors affecting conductance of a solution. Write down the applicationsofconductometry in brief.	05